WATERMARKING TECHNIQUES FOR DISTRIBUTED DATABASES

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THESIS CERTIFICATE

This is to certify that the thesis titled **WATERMARKING TECHNIQUES FOR DISTRIBUTED DATABASES**, submitted by **Yashwanth Duvvuru** , to the Indian Institute of Technology, Patna, for the award of the degree of **Bachelor of Technology**, is a bonafide record of the research work done by him under our supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

KEYWORDS: Watermarking,Databases,Relational,Distributed.

Now a days,enormous amount of data is being generate each day.It is increasingly important to ensure the integrity of these data ,ownership proof of the data and prevent illegeal reselling.Digital watermarking technology was suggested in the last decade to embed copyright information in digital objects such images, audio and video.A Lot of watermarking techniques have been developed for relational centralized databases also.However,the increasing use of distributed database systems in real-life applications created an increasing need for watermarking distributed database systems.In this thesis,we propose a new watermarking technique for distributed database system which describes how to watermark the database before it is distributed.In this technique the data owner considers the partition overview from the third party and watermarks the data according to the partition overview by the third party.The most crucial point is that the watermark is partition independent.

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CHAPTER 1

INTRODUCTION

Now a days, distribution of digital data has become more sophisticated due to the development of various new Internet techniques. This is creating pressure on data providers to create services that allows users to access the databases remotely.The piracy of digital data like software, images, video, audio etc has become a concern for owners of these digital assets. Protection schemes are usually based upon the insertion of digital watermarks into the data.Watermarking is nothing but introducing small errors into the object being watermarked. These intentional errors are called marks, and all the marks together constitute the watermark.Watermarking doesnot prevent copying,but it deters illegal copying by providing a means to establish ownership of a copy. A new technology, digital watermarking, provides a promising method of protecting digital data from illicit copying and manipulation by embedding a secret code directly into the data.

Watermarking is the process of hiding digital information in a carrier signal; the hidden information should, but does not need to contain a relation to the carrier signal. Watermark has wide range of applications like tamper detection, localization, ownership proof, traitor tracing purposes etc .In general, the database watermarking techniques consist of two phases: Watermark Embedding and Watermark Verification.During insertion phase,a key is used to embed watermark information into an original database.This produces watermarked database which can be later used for publication or distribution.With appropriate key and watermark information,a watermark detection process can be applied to an suspicious database to verify the ownership of the database. Figure 1.1 shows process of embedding phase and figure 1.2 shows process of detection phase.

Initially, most of the work on watermarking was concentrated on watermarking of still images, video, audio etc. However, in the last few years Watermarking of database systems started to receive attention because digital data can be easily duplicated and modified. There is a great deal of concern about the integrity and intellectual property protection of these data. Database Watermarking is one technique which can overcome

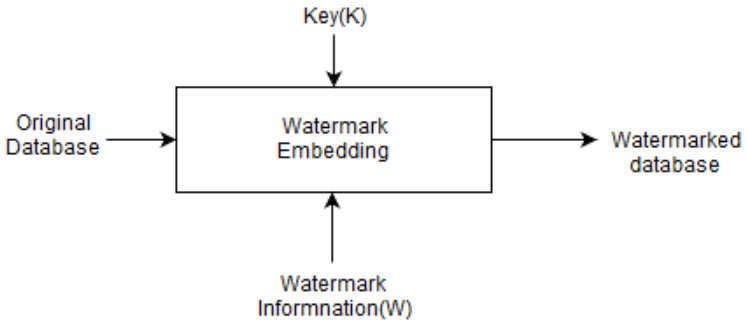

Figure 1.1: Watermark Embedding

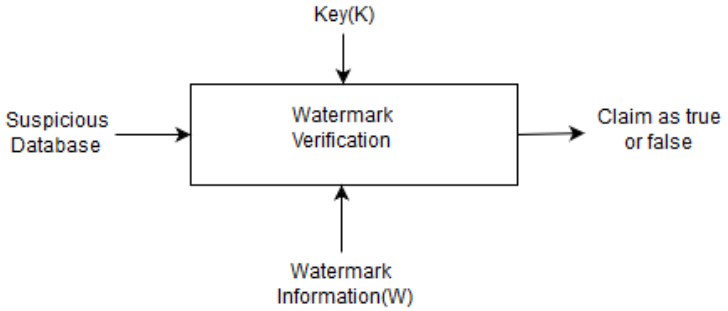

Figure 1.2: Watermark Verification

the problem of copyrights, false ownership issues; detection of the tampering in the databases.

Now a days,most of the organisations are looking to outsource their data so that they can provide their services to the users remotely.For this purpose,they have to distribute their data among their servers at various places.But there are so many challenges and security issues that the data owners need to overcome to handle this whole process smoothly.So there is a demand for technique which solves the security issues involved in the distributed systems and also which identifies the pirated copies of the original databases.

Examples where database watermarking might be of a crucial importance include protecting rights and ensuring the integrity of outsourced relational databases. in service provider model , in data mining technology where data are sold in pieces to parties specialized in mining it, online B2B interactions etc.

1.1 Motivation

In recent time, there is a need to outsource the database by the data owners to any third party that can manage the database. Distributed system is always an attractive solution as a cost effective and economic way to manage this huge data.

There are several papers proposed on relational database watermarking , but there is not any proper technique for watermarking the distributed database. Significant work has not been done in this area till date. Only a single patent is available that aims to watermark the database in distributed environment. The watermarking method described in this patent, takes the whole database as input to insert watermark that may lead to various issues like illegal reselling, ownership claims etc. Also at the time of detection, the data owner has to extract all the watermarks from top to bottom in case of multilevel watermarking. Major drawback in this patent is that, they have not considered any criteria of partitioning of the database. This motivated me to propose a new watermarking technique for distributed database system.

1.2 Structure of Thesis

In this thesis,Chapter 2 describes about related work done in the field of watermarking.Chapter 3 describes about distributed databases.In Chapter 4,the proposed technique has been described and Experimental results have been tabulated.

CHAPTER 2

Related Work

In this chapter, classification of watermarking techniques ,properties of watermarking techniques ,various types of attacks and prior works on watermarking have been discussed.

2.1 Difference between Steganography and watermarking

Steganography is the art and science of hiding information in ways that prevent the detection of hidden messages. Steganography literally means âĂIJcovered writingâĂİ and is usually interpreted to mean hiding information in other information.

Watermarking is the process of embedding a message on a host signal. Watermarking, as opposed to steganography, has the additional requirement of robustness against possible attacks. A watermark can be either visible or invisible.

In the [1] the author stated some differences between Steganography and Watermarking.They are :

(i)The information hidden by a watermarking system is always associated to the digital object to be protected or while steganographic systems just hide any information

(ii)Robustness criteria are also different, since steganography is mainly concerned with detection of the hidden message while watermarking concerns potential removal by a pirate.

(iii)Steganographic communications are usually point-to-point while watermarking techniques are usually one-to-many.

2.2 Various Types of Watermarking

The digital watermarks can be divided into two types based on the visibility of the watermark as follows:

(i)Visible watermarking: As the name suggests, visible watermarking refers to the information visible on the image or video or picture. Visible watermarks are typically logos or text. For example, in a TV broadcast, the logo of the broadcaster is visible at the right side of the screen.

(ii)Invisible watermarking: Invisible Watermarking refers to adding information in a video or picture or audio as digital data. It is not visible or perceivable, but it can be detected by different means. It may also be a form or type of steganography and is used for widespread use. It can be retrieved easily.

Here is another classification of watermarking techniques based on public key and secret key.

(i)Public watermarking: In public watermarking, a public key which is used for detecting watermark is available publicly. Only the copyright holder has his private key to embed a watermark. Anybody who knows the public key could detect the watermark. Therefore, public-key watermarking can provide public detection, but the secret key cannot be deduced from the public key. Also, knowing the public key does not enable an attacker to remove the watermark.

(ii)Private watermarking: In case of private watermarking techniques ,the watermarking is based on private parameters i.e. Secret Key.Most of the existing watermarking techniques are private.Only the authorized people (e.g. database owners) who know these private parameters are able to verify the watermark and prove their ownership of the database in case of any illegal redistribution, false ownership claim etc.

Based on whether the marking introduces any changes in the underlying data of the database, the watermarking techniques can be categorized into two:

(i)Distortion based watermarking: In this type of watermarking,during the embedding phase,small changes are made to the original data of the database.The changes made should be tolerable and should not make the data useless. As mentioned already,The watermarking in this category is generally done at bit level,character level,attribute

or tuple level or even on the values in tuples and attributes .

(ii)Distortion free watermarking: In contrary to the distortion based watermarking ,distortion free watermarking techniques donot make any changes to the underlying data. Most of the distortion free techniques are fragile.The distortion watermarking techniques aim at maintaining the integrity of the database in addition with ownership claiming.This type of watermarking donot depend on any specific type of attribute.

2.3 Differences between Multimedia watermarking and database watermarking

In [2] the differences between multimedia watermarking and databse watermarking are clearly stated.It states that is reasonable to apply the basic ideas of multimedia digital watermarking algorithms in database watermarking techniques but we can not directly apply them.Watermarking of relational data defers from watermarking of multimedia data in many respects.Some of them are:

(i)Few Redundant Data: Multimedia data objects usually consist of large numbers of bits with redundancy in which plenty of digital watermarking information can be stored. Since every space of tuple stores identified value leading to low redundancy, it is not easy to store watermarking information in separate tuples dispersedly.

(ii)Out of Order Relational Data: Each point in multimedia data objects has constant relative space and time positions, while in relational database tuples and fields to form tuples are not displayed in specific sequence.

(iii)Frequent Updating: Multimedia data stay unchanged in most conditions, where as it is necessary to update and maintain data in relational database.

(iv)Detectability: If there are some changes or missing data in certain parts of multimedia data objects, they are easily detectable. In relational database, even partial tuples are replaced or deleted generating changes of database they are hard to detect.

2.4 Properties of database watermarking

As stated in [3] there are four main Properties of digital watermarking technique which are to be considered while watermarking databases.They are:

(i)Transparency or Fidelity: The applied watermark should not affect the quality of the original data after it is watermarked. Watermarking should not introduce visible distortions because if such distortions are introduced it reduces the commercial value of the data.

(ii)Robustness: The capability of the watermarking scheme to survive deliberate changes like modifying, adding, deleting part of the data and unintentional attacks like digital reproduction and photocopying. The watermark should be detectable even in an object modified by the attacker.

(iii)Capacity or Data Payload: This property describes how much information should be embedded as a watermark to successfully detect during extraction. Watermark should be able to carry enough information to represent the uniqueness of the data. Different application has different payload requirements .

(iv)Detectability: The owner of the database should be able to detect the watermark by examining the tuples from the suspicious database.

//write about various types of attacks.

2.5 Different types of Attacks

After watermarking,the watermark may suffer from unintentional attacks or intentional attacks from the attackers which may cause damage to the watermark.Some of these attacks are mentioned below:

1.Benign Update:In this case,the data of the watermarked relation may be processed as usual.In this type of attack,the marked tuples are removed,added or updated which may cause damage to the watermark and make the watermark undetectable.

2.Value modification Attack. In this type of attack,the attacker attempts to destroy the watermarked data by flipping the bits of watermarked data.The extent of damage

depends on how many bits are exposed to attacker.Bit attack,Rounding attack are examples of value modification attack.

3.SubSet attack. The attacker considers subset of attributes or subset of tuples from the watermarked database and attacks them which may cause damage to the watermark.

4.Superset attack. The attacker adds some new tuples or attributes to the watermarked database which damages the watermark and may affect the correct detection of watermark.

5.Additive attack:In this type of attack,the attacker adds his own watermark to the watermarked database and claims the ownership of that database.

2.6 Prior works on watermarking

The watermarking scheme proposed by Agarwal and cortesi (popularly known as AHK algorithm) in [4] is based on numeric type attribute and watermarking is done at bit level.The basic idea is some bits of the attributes of the some of the tuples are which are selected using hash function are marked .The bit pattern constitutes the watermark for the database.r.A_i denotes the value of attribute A_i in the tuple r belongs to R. The selection of bit postions depends on n,V,E,K parameters specified above. The Hash function H(K|| H(K||r.P)) is used to derive the positions at which mark is to be embedded,where r.P denotes the primary key of tuple r and || denotes concatenation function.Similarly the Hash function H(K||r.P) computes the value to be embedded at the derived positions.

The following algorithm proposed in [5] which is a modification of AHK algorithm also embeds the watermark at bit level in a reversible manner.In this scheme,during detection,along with ownership proof,original watermark can also be recovered. The operation involves extracting bit oldBit from the integer portion of the attribute before replacing it by the watermark bit and inserting it in the fraction portion of the attribute.Thus during detection phase,we extract the Oldbit from the fraction part and keep the bit back at its original position which restores the original database.

The watermarking algorithm proposed in [6] are directly performed on the content of the database itself. In this scheme, we select two attributes local attribute and characteristic attribute.For a generated random number a which is less than embedded propor-

tion (a) of the relational database , if the value of the characteristic attribute A_1 is not null, we can watermark the tuple.A local characteristic is extracted from the characteristic attribute A_1 of a particular tuple in defined order.The extracted bits are embedded into the watermark attribute A_2 of the same tuple from which bits are extracted.The watermark detection phase is almost a reversible process. we extract the local characteristic of the characteristic attribute and last bits of the watermark attribute and check the both patterns.

The follwoing watermarking scheme is proposed in [5].In this scheme,all tuples are first divided into groups according to the number of groups g, the hash value of a embedding key K, and their primary key. The tuples in each group are then sorted based on their primary key. watermarks are embedded into each group independently. In each group, there are two kinds of watermarks: attribute watermark W_1 which consists of c watermarks of length m and tuple watermark W_2 which consists of m watermarks of length c. the embedding subroutine mainly consists of two parts: attribute watermark embedding and tuple watermark embedding. For attribute watermarks, the hash value is generated according to a message authentication code and the same attribute of all tuples in the group, while for tuple watermarks, it is formed according to the same message authentication code and all attribute values of the same tuple.

The following watermarking scheme proposed in [7] embeds a meaningful watermark information by first converting it into a bit flow. The scheme computes unique ID for all tuples in the relation and sort them in ascending order according to their ID values. The tuples are then partitioned into p groups each containing m tuples. The ith bit of the bit flow is embedded into the selected tuples in i^th group by following same selection criteria as in AHK algorithm with exception that it considers only single attribute to mark. Before committing the change, a constraint function is used to check whether the change exceeds the data usability bounds.

The following approach proposed in [8] is probabilistic and rather uses probability distributions to mark the data.The data owner decides number of fake tuples to be generated.The watermark insertion algorithm uses two values p_i and F_i where they indicate sensitivity of the element a_i attribute from A_i (0< p_i <1) and set of fake values for attribute a_i respectively.Depending on the value of p_i ,we decide whether to select a value from set of fake values uniformly or using Biased Sampler.Unlike other algorithms, the

detection algorithm is not an inverse algorithm to the watermark generating algorithm .The detection algorithm checks the fake tuples using primary key whether they been updated .As soon as one match is found ,the detection is done.If the user deletes all the fake tuples,it is impossible to detect the watermark through detecting algorithm.

The following scheme proposed in [9] inserts virtual attribute into the relation rather than fake tuples.The data is first portioned in number of non-overlapping sets.The virtual attribute serves as the watermark for this data.The values of the virtual attribute are calculated parity checksum of all other attributes and an aggregate value obtained from any one of the numeric attribute of all tuples..Watermark detection algorithm is a reverse process for watermark insertion.

In [] a new distortion-free technique was proposed.In this technique , all tuples are divided into g groups according to a a secure primary key hash value is computed for each tuple according to an embedding key and the primary key the tuple. The grouping does not change the physical position of the tuples. After grouping, all tuples in each group are sorted according to their primary key hash. Before embedding, a secure tuple hash is computed for each tuple based on the embedding key and all attributes of the tuple. A group hash value is then computed based on the tuple hash of sorted tu- ples in the group and the embedding key.To embed the watermark, for each tuple pair, the order of the two tuples are changed or unchanged according to their tuple hash values and the corresponding watermark bit. As we can see, since only the order of tuples is changed, the watermark embedded is distortion free.

CHAPTER 3

Proposed Watermarking technique for Distributed Databases

A distributed database is a single logical database that is spread physically across computers in multiple locations that are connected by a data communications network..The network must allow the users to share the data; thus a user(or program) at location A must be able to access (and perhaps update) data at location B.

A decentralized database is also stored on computers at multiple locations; however, the computers are not interconnected by network and database software that make the data appear to be in one logical database. Thus, users at the various sites cannot share data. A decentralized database is best regarded as a collection of independent database

3.1 Advantages

Compared with centralized databases, either form of a distributed database has numerous advantages. The following are the most important of them:

(i)Increased reliability and availability: When a centralized system fails, the database is unavailable to all users.Where as in case of distributed database, distributed system will continue to function at some reduced level, however, even when a component fails. The reliability and availability will depend on the way the data are distributed

(ii)Local control: Distributing the data encourages local groups to exercise greater control over their data, which promotes improved data integrity and administration.

(iii)Modular growth: In a distributed environment, it is much easier to handle expansion. New sites can be added to the network without affecting the operations of other sites. This flexibility allows an organization to expand relatively easily. Adding processing and storage power to the network can usually handle the increase in database size.

(iv)Lower communication costs With a distributed system, data can be located closer to their point of use. This can reduce communication costs, compared with a central system.

(v)Faster response: Most requests for data by users at a particular site can be satisfied by data stored at that site. This speeds up query processing since communication and central computer delays are minimized. It may also be possible to split complex queries into subqueries that can be processed in parallel at several sites, providing even faster response.

3.2 Challenges

Even though many advantages are present for distributed databses,there exists some challenges in establishing distributed systems.Here are some challenges:

(i)Software cost and complexity: A distributed DBMS that hides the distributed nature from the user and provides an acceptable level of performance, reliability, availability is inherently more complex then a centralized DBMS. If the software does not handle data replication adequately, there wi1l be degradation in availability, reliability and performance compared with the centralized system

(ii)Data integrity: Database integrity refers to the validity and consistency of stored data. Integrity is usually expressed in terms of constraints, which are consistency rules that the database is not permitted to violate. Enforcing integrity constraints generally requires access to a large amount of data that defines the constraints.

(iii)Processing overhead: The various sites must exchange messages and perform additional calculations to ensure proper coordination among data at the different sites.

(iv)Slow response: It the data are not distributed properly and the query is not formulated correctly, it may result into significant slower response.

3.3 Partitioning

Distributed databases are partitioned and then they are made available for services.There are four different strategies for partitioning the data.

(i) **Data Replication :** Data replication means storing the separate copy of the database at each of two or more sites.All the servers in the network have same database.This approach is favoured when most process requests are read-only requests.This approach is not appropriate for distributed databases and impractical except for only relatively small databases.

(ii)**Horizontal partitioning:** In general,in horizontal partitioning,the rows of a relation are distributed to many sites.This approach is more suitable when an organizational function is distributed but each site is concerned with only a subset of entity instances.

(iii) **Vertical partitioning:** In general ,in vertical partitioning,the columns are projected to several sites.To reconstruct the table partitioned by vertical partitioning,the relations at each of the sites must share a common domain.Reconstructing of table in vertical partitioning is difficult when compared to horizantal partitioning.

(iv)**Hybrid Partitioning:** Any combination of the above approaches is called hybrid partitioning.There can be instances where both horizantal and vertical partitioning are desirable.In such cases,hybrid partitioning is used to distribute the data.

3.4 Proposed Technique

We propose a novel watermarking technique for distributed database.Our consideration is that data owner does not have enough resources to manage the data and the data owner outsources the data to the third party considering that the third party is capable of managing data. Some of the challenges here are: (i) Distribution of data (ii) Existence of replication (iii) Preservation of watermarks while performing partition and distribution by third party.(iv) Key management.

There are three different phases in the proposed technique.Three phases are:

1.Initial exchange of Partition information-The third party provides the informa-

tion about partitioning and distribution i.e. Partition overview (ψ) upon data owner's request.The data owner provides only the schema of the database to the third party at this phase.

2. Watermark Embedding by data owner- There are two main steps in this phase : Key management and Watermark embedding.

(a)Based on the partition overview (ψ) received from the third party, the data owner has to watermark the database.The data owner selects a set of n different keys obtained from a secret key K where n represents the possible number of fragments obtained from the partition overview ψ.

(b)Using this information the data owner watermarks the database with different keys satisfying the partition overview. The data owner performs the watermarking as WM(DB, W, K, π) = DB_w where K = {k_i|i $\in$I } .

(c)Data owner now outsources the watermarked database DB_w to the third party.

3. Partitioning and Distribution by third party- Finally, the third party will partition the watermarked database considering the partition overview used in the first step. After partitioning ,the third party distributes the partitioned database and stores in various servers at different locations.The third party maintains the metadata table to store all these information about servers and the partition residing on these servers.

$F_{part}(DB_w,\psi)$ = { MD, DB_{w_i} }.

After distributing the partitions ,if the data onwer finds any suspicious data ,then the data owner runs the detection algorithm and verifies the ownership of the suspicious data.This technique consists of both the embedding and verification phases of database watermarking .By using this technique,there would not be any damage to watermark even after partitioning.

CHAPTER 4

IMPLEMENTATION

4.1 Experimental Analysis

We have performed our experiment using the Forest Cover Type data set 1. The data set has 581012 tuples, each having 61 attributes. An extra attribute id is added in our experiment that serves as primary key. We successfully implemented Embedding , partitioning and detection for various number of partitions.

4.2 Key Management

As our proposed technique is for distributed databases,it is very importat to select the private key K and watermark the database using key K such that watermark is partition independent.There are four different scenarios to consider in this direction.They are:
(i) same watermark, same key
(ii)same key, different watermark.
(iii) different key, same watermark
(iv) different key, different watermark.

In the first two cases,the watemark ha high chances of being vulnerable because if at any one site, watermark is revealed,then the watermark at the other sites can be determined very easily.In the last case applying different watermarks along with different keys will be a tedious job. Therefore we have considered the third scenario which is the most appropriate one.In the third case,even if watermark is revealed at one site,it is difficult to determine the watermark at other sites. This also serves the purpose of partition-independent watermarking. To implement this scenario ,we have considered the secret sharing schemes.The reason is that by using the secret sharing schemes managing and distributing large number of independent keys becomes easier.

Secret sharing consists of recovering a secret S from a set of shares, each containing partial information about the secret. There are several types of secret sharing schemes. The most basic types are called threshold schemes, where only the cardinality of the set of shares matters. In other words, given a secret S, and n shares, any set of t shares is a set with the smallest cardinality from which the secret can be recovered, in the sense that any set of t-1 shares is not enough to give S.Key Management scheme used for this implementation is k out of n secret sharing.

Various k out of n secret sharing schemes are already proposed previously such as: Shamir's scheme [10], Mignotte's scheme [11], etc. However in Shamir's scheme, if the attacker knows even k - 1 keys, he will eventually get a range of numbers to guess about the secret key .Therefore we have used Mignotte's scheme that overcomes the drawback of Shamir's scheme.

4.3 Watermark Embedding

Symbol	Description
Count	no. of tuples used for particular experiment
ν	no. of attributes used for marking and detection in the relation
γ	fraction of tuples used in the experiment
χ	no. of least significant bit available for marking in an attribute.
TC	total count that is marked during embedding.

Table 4.1: Notations

The data owner watermarks the database with different keys obtained through Mignottes scheme and also considering the partition overview obtained from the thrid party.

Data owner performs the watermarking as Distributed WM(DB, W, K) = DB_w = $\bigcup DB_w$ i where i =1, 2, ..., n and n is the possible number of partitions in ψ. DB_i is the particular fragment which is to be watermarked with key K_i obtained from secret key K by using Mignotte's secret sharing scheme. For the purpose of watermarking, we implemented AHK algorithm [12] to watermark the tuples.

During embedding,each tuple is selected at first and and checks whether it satisfies the condition to embed the watermark.If the tuple satisifies the condition,the mark is embedded in one of the selected attributes which is calculated using attribute index.In the selected attribute the watermark is to be embedded at some specific bit positions which are calculated sing bit index.

Count	ν	χ	γ	TC	time(in ms)
581012	10	15	20	28942	14871269
581012	5	15	20	57884	33262117

Table 4.2: Experimental Results for γ=20.

Count	ν	χ	γ	TC	time(in ms)
581012	10	15	50	11851	14213425
581012	5	15	50	23702	27380395

Table 4.3: Experimental Results for γ=50.

4.4 Partitioning

The third party can partition the watermarked database dbw according to the partition overview provided by them to the data ownwr in the previous step.After partitioning ,the third party maintains a database table named metadata to store partition and corresponding server in which partition is stored.As already mentioned the partitioning can be either horizantal or vertical or hybrid partitioning .After each partitioning,partition ${DB_w}^i$ is send to sever S_j and (${DB_w}^i$,S_j) is stored in metadata. The schema of the metadata table consists of Server-id(S_j),Schema of the partition stored in that particular server and properties of that partition.For example the tuple in the metadata table would be like:$\{S_j, < \{A_1, A_2, A_3\}; A_3 \leq AVG(A_3) >\}$

The experimental results regarding number of tuples in each partition with respect to number of partitions are shown in the following tables.For all the cases the value of γ is considered as 50.

Partition	Number of tuples
P1	337195
P2	243817

Table 4.4: Number of tuples in each partition for 2 partitions.

Partition	Number of tuples
P1	337195
P2	243817
P3	280692
P4	300050

Table 4.5: Number of tuples in each partition for 4 partitions.

Partition	Number of tuples
P1	330969
P2	250043
P3	337195
P4	243817
P5	336456
P6	245366
P7	246035
P8	334977

Table 4.6: Number of tuples in each partition for 8 partitions.

4.5 Detection

This is one of the challenging steps in the whole process.If data owner finds any suspicious or pirated database,then the data owner applies the detection algorithm and verifies the ownership of the database.The steps involved in the detection process are:

1.If the data onwer finds the suspicious database,then data owner contacts the third party for the server-id of the server on which the suspicious database is residing.

2. To find out the server-id of the susupicious database,the third party refers to the metadata and provides the server-id and partitioned database to the data owner.

3.Using the Private Key K and server-id obtained from the third party,the data owner finds the key K_i which was used to watermark that partition $DB_w i$.

4.Using the key calculated in the above step,i.e. K_i the data owner can extract the watermark W and verify the ownership of the database.

We have applied this detection algorithm to the forest data set as mentioned earlier.In this scenario we considered attacker updates the database to certain extent.So after updation by the attacker,we run the detection algorithm to verify the onwership of the database. We have considered three scenarios on how many tuples will be updated by the attacker.In first scenario,the attacker doesnot change anything,in second scenario 50% and in third scenario the attacker updates 90%.In the third scenario,we considered two different cases which differ in the value of ξ.We have considered these three scenarios for three different number of partitions.

In this paragaraph ,the procedure for calculation of threshold is mentioned.Suppose

at the time of embedding the data owner marks a total of ω bits.Now during the detection process the data owner observes number of bits matches the values assigned by the marking algorithm.The probability that atleast τ bits out of ω bits which have equal probability for being 0 or 1 , matches the assigned values is $B(\tau;\omega,1/2)$,where B denotes Bernoullis probability.Now the minimum value of τ is selected in such a way that it satisfies $B(\tau;\omega,1/2) \leq \alpha$,where α denotes significance level.α is the probability that the data owner will discover the watermark in a database relation marked by her.

The detectability of a watermark depends on the significance level α and number of marked tuples ω.And ω depends on number of tuples in relation and fraction of tuples marked(χ).

The following table shows the experimental results for detection algorithm for 2 partitions.In all the scenarios the updation of the attacker is successfully detected by verifying the watermark.The attack on each partition is considered independently.τ is calculated for each case using significant level α . Throughout the detection algorithm,the value of γ is taken as 50.

Partition	**Updation**		**Detection**			
count	percent updated	χ-bit updated	match count	τ	time (msec)	Detect
337195	0	NA	6791	3395	4610860	✓
	50	8	6341	3395	4104676	✓
	90	8	5977	3395	4247292	✓
		10	4741	3395	4395566	✓
243817	0	NA	5060	2530	2668006	✓
	50	8	4682	2530	2239336	✓
	90	8	4449	2530	2495227	✓
		10	3525	2530	2303253	✓

Table 4.7: Detection after partitioning and updation in case of 2 partitions

In table 4.7,if match count is greater than threshold(τ),then the watermark in the database is successfully detected.In all the cases, τ is less than match count.So in this scenario,all the cases are successfully detected.

The following tables shows the experimental results for detection algorithm for 4 partitions and 8 partitions respectively.In these two cases,the same scenarios consideres for 2 partitions are considered and detection algorithm is implemented for these scenarios.In all the scenarios the updation of the attacker is successfully detected by verifying the watermark.The attack on each partition is considered independently. Throughout the detection algorithm,the value of γ is taken as 50.

Partition	Updation		Detection			
count	percent updated	χ-bit updated	match count	τ	time (msec)	Detect
337195	0	NA	6791	3395	3086946	✓
	50	8	6317	3395	3148158	✓
	90	8	5988	3395	3248881	✓
		10	4759	3395	3052083	✓
243817	0	NA	5060	2530	1609398	✓
	50	8	4618	2530	1668649	✓
	90	8	4446	2530	1677063	✓
		10	3535	2530	1677395	✓
280962	0	NA	5753	2876	2192104	✓
	50	8	5509	2876	2099414	✓
	90	8	5320	2876	2208180	✓
		10	3990	2876	2198221	✓
300050	0	NA	6098	3049	2477041	✓
	50	8	5824	3049	2630520	✓
	90	8	5629	3049	2501038	✓
		10	4307	3049	2473426	✓

Table 4.8: Detection after partitioning and updation in case of 4 partitions

Partition	Updation		Detection			
count	percent updated	χ-bit updated	match count	τ	time (msec)	Detect
330969	0	NA	6721	3360	2781232	✓
	50	8	6205	3360	2699055	✓
	90	8	5887	3360	2650933	✓
		10	4678	3360	2630398	✓
250043	0	NA	5130	2565	1600756	✓
	50	8	4756	2565	1568929	✓
	90	8	4545	2565	1528359	✓
		10	3612	2565	1523670	✓
337195	0	NA	6791	3395	2790800	✓
	50	8	6791	3395	2795596	✓
	90	8	6791	3395	2768237	✓
		10	4761	3395	2775518	✓
243817	0	NA	5060	2530	1523984	✓
	50	8	5060	2530	1466059	✓
	90	8	5060	2530	1471725	✓
		10	3541	2530	1490088	✓
335646	0	NA	6833	3416	2836748	✓
	50	8	6833	3416	2791317	✓
	90	8	6833	3416	2809398	✓
		10	4804	3416	2765616	✓
245366	0	NA	5018	2509	1484451	✓
	50	8	5018	2509	1465203	✓
	90	8	5018	2509	1508419	✓
		10	3489	2509	1480657	✓
246035	0	NA	4956	2478	1516136	✓
	50	8	4956	2478	1475345	✓
	90	8	4956	2478	1514514	✓
		10	3469	2478	1526597	✓
334977	0	NA	6895	3447	2676237	✓
	50	8	6895	3447	2705557	✓
	90	8	6895	3447	2817721	✓
		10	4830	3447	2720335	✓

Table 4.9: Detection after partitioning and updation in case of 8 partitions

4.6 Experimental Observations:

After performing the experiments for three different number of fragments and different values of χ,Some observations were found.They are:

1.With the increase in the value if γ the value of Total count decreases.

2.With the decrease in the value of γ,the robustness of the watermark increases because number of marks increases.

3.With the increase in the value of number of attributes available for marking,the robustness increases because number of marks increases.

4.With the increase in the percentage of updation by attacker,the macth count values decreases and approaches threshold.If we keep increasing the percentage of updation then at one stage Totalcount goes below the threshold and watermark cannot be detected.

5.For same percentage of updation,if the value of χ increases then the value of match count decreases and approaches the threshold value.

6.With the increase of count i.e. number of tuples in the partition ,the threshold value (τ) increases and probability of successful detection decreases.

4.7 Summary

The major contribution of this thesis is development of new watermarking technique for distributed databases .The thesis also provides detailed study about issues and challaneges faced while outsourcing the data to the third party.This technique is successfully implemented on a real life database and successful results were yielded.Detection of watermark is successful for various number of partitions.

REFERENCES

[1] S. Katzenbeisser and F. Petitcolas, *Information hiding techniques for steganography and digital watermarking*. Artech house, 2000.

[2] R. Halder, S. Pal, and A. Cortesi, "Watermarking techniques for relational databases: Survey, classification and comparison." *J. UCS*, vol. 16, no. 21, pp. 3164–3190, 2010.

[3] I. J. Cox, M. L. Miller, and J. A. Bloom, "Watermarking applications and their properties," in *itcc*. IEEE, 2000, p. 6.

[4] R. Agrawal and J. Kiernan, "Watermarking relational databases," in *Proceedings of the 28th international conference on Very Large Data Bases*. VLDB Endowment, 2002, pp. 155–166.

[5] H. Guo, Y. Li, A. Liu, and S. Jajodia, "A fragile watermarking scheme for detecting malicious modifications of database relations," *Information Sciences*, vol. 176, no. 10, pp. 1350–1378, 2006.

[6] Y. Zhang, X. Niu, D. Zhao, J. Li, and S. Liu, "Relational databases watermark technique based on content characteristic," in *Innovative Computing, Information and Control, 2006. ICICIC'06. First International Conference on*, vol. 3. IEEE, 2006, pp. 677–680.

[7] M. Huang, J. Cao, Z. Peng, and Y. Fang, "A new watermark mechanism for relational data," in *null*. IEEE, 2004, pp. 946–950.

[8] V. Pournaghshband, "A new watermarking approach for relational data," in *Proceedings of the 46th Annual Southeast Regional Conference on XX*. ACM, 2008, pp. 127–131.

[9] V. Prasannakumari, "A robust tamperproof watermarking for data integrity in relational databases," *Research Journal of Information Technology*, vol. 1, no. 3, pp. 115–121, 2009.

[10] A. Shamir, "How to share a secret," *Communications of the ACM*, vol. 22, no. 11, pp. 612–613, 1979.

[11] M. Mignotte, "How to share a secret," in *Cryptography*. Springer, 1982, pp. 371–375.

[12] R. Agrawal, P. J. Haas, and J. Kiernan, "Watermarking relational data: framework, algorithms and analysis," *The VLDB journal*, vol. 12, no. 2, pp. 157–169, 2003.